Summer Project: Project Proposal

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Abstract—Welcome to the project proposal for the summer project I had in mind! Here I display my ideas on how I integrate different skills and niche together to create, what I believe, an impressive project!

Keywords—*Summer Project, Fun, Independent Project, Experience*

1. Proposal Introduction

Imagine computer vision, mechanical engineering, and human-computer interaction, all coming together seamlessly in one revolutionary project. I'm excited to introduce: **Your Remote Arm**–a prototype and concept born from my belief that the most **impactful** innovations happen when advanced technology meets a intuitive, user-oriented design.

Your Remote Arm isn't just another tech gadget; it's a groundbreaking machine that systematically **evolves** to perfectly replicate tasks. Our idea is to create a mechanical arm that can replicate a user's arm movements over a period of time. This replication is taken in the form of a video feed–in any form (e.g. Youtube, Webcam, iPhone, etc.)–and could be stored and reused with minimal data loss. This storing system would be in the form of a large platform where users can easily repeat tasks that were previously uploaded. Behind the scenes, a perfect blend of control theory and computer vision optimizes every movement, ensuring the robotic arm performs with the same finesse as your own. The project model is designed to be user-centered and **applicable to every field in any industry**. From precise surgeries, to meticulous circuit boards, and even to finance restrictions, this project will include it all. This project is designed to be both challenging and transformative, providing a roadmap that is documentable, achievable, and, most importantly, **educational**. I envision a collaborative journey where every team member grows alongside the project, gaining deep insights into both the cutting-edge technology and the interdisciplinary process behind it.

2. Our Goal

Our goal is to create a machine that can replicate physical tasks with minimal use of sensors and manual interference. My goal is to facilitate the growth and success of my team so that we not only complete the first milestone by the end of the summer but also have the knowledge and experience to present all parts of this project.

3. Key Features

This project, while absolutely expandable, will be limited to **Mechanical Engineering**, **Computer Vision** and **Optimization** due to the time constraints of the summer. Here are the following features I've deemed important to relay.

3.1. Robotic Arm

The physical product will be a robotic arm. This will most likely be something bought off-the-shelf to ensure that we have a functional arm with functional software. Our role is to design the hand/claw to be adaptable to different tasks. This may be in the form of multiple easily-installed claws or a singular one. For this summer, our goal is to successfully replicate single-hand precise tasks such as soldering or even origami (assume a controlled task that can be accomplished using a single arm). The data will initially be in the form of a controlled recording setting and will expand the further we get into this project.

3.2. User Platform

This will most likely not get accomplished over the summer. Other than optimizing data storage and reconstruction, the actual interface will most likely not be covered this summer. This platform will be most tricky in that it involves modularizing a seamless transfer of data from an uncontrolled video feed, to mechanical instructions, to a **cloud-based platform**, and finally to be repeatedly used in a live setting. The cloud-based platform must also allow easy navigation for the users and potentially a feature to modify existing machine instructions.

4. Key Checkpoints and Timeline

Considering many proposed features will not reach the fully deployment-ready stage by the end of the summer, and that some team members will have other commitments, the following milestones are designed to be achievable within the allotted time. This summer, it is also my goal to have each member contribute to each task even if it is out of their expertise. Being able to absorb, process, and reflect information will, evidently, be a key trait I look at when recruiting members.

4.1. Summer Milestone - Sync Human and Computer

We must develop and execute optimal techniques in order to get the highest precision in syncing user movements with the arm.
Features Required: Control Theory, Optimization, Computer Vision, System Integration, and Modularity
Skills Required: Mechanical/Electrical Engineering, Data Structures and Algorithms, System Communication, **Practical** Experience in Optimization and Computer Vision

4.2. Second Milestone - Data reconstruction

If we still have time, we must attempt to compress and decompress the data–currently in the form of mechanical instructions–such that the reconstructed task can be replicated with a high certainty.
Features Required: Data storage, compression, reconstruction, and retrieval
Skills Required: Storage Systems, Practical Optimization, System Communication

4.3. Third Milestone - Platform and Interface

We must create the platform and develop the interface that will store the data.
Features Required: User-oriented Platform, Visual Theory
Skills Required: Front-End Programming, System Communication

4.4. Fourth Milestone - Modularization and Integration

We must modularize all the robot parameters and integrate them to the platform.
Features Required: Modularization
Skills Required: System Communication

4.5. Minor Milestones

We must deploy a prototype to get advice and practical feedback about the model. Following, we must take action to the feedback and integrate it to the system. If all goes well, I will look for opportunities to showcase the finished product.

5. Conclusion

This project embodies the heart of human-computer interaction–the idea that technology should empower people, adapt to their needs, and enhance their everyday lives. By allowing users to program and refine the system themselves, Your Remote Arm not only pushes the boundaries of what's possible in robotics but also invites a community-driven approach to innovation.